# NovaMart Intelligence

Enterprise KPI Intelligence-to-Action Engine — Official Technical Documentation & Architecture Specification

Live Production: https://kpi-engine.vercel.app | GitHub: Vapa1510/accenture_businessintelli | Accenture Hackathon Track 3 | Strict Quant/LLM Split

## 1. Executive Summary & Core Principle

**NovaMart Intelligence** is an enterprise-grade analytics engine designed to detect material KPI movements across e-commerce operations, explain their root causes using multi-source evidence, and recommend actionable operational interventions — or issue an explicit **selective abstention** when available data is stale, sparse, or contradictory.

> **The Core Technical Pillar: "Never Let the LLM Be the Source of Numbers"**
>
> Every financial figure, percentage change, regression coefficient, factor decomposition, and confidence score is computed **100% deterministically in Python** server-side. The Large Language Model (LLM) is restricted to prose formatting and is governed by post-hoc Pydantic numerical verification before any response reaches the user.

## 2. Live Production & Setup Specification

**LIVE PRODUCTION URL**

**https://kpi-engine.vercel.app**

Deployed on Vercel Serverless (FastAPI + React Vite)

**GITHUB REPOSITORY**

**Vapa1510/accenture_businessintelli**

Full open-source code, test suite & documentation

### Quick Start Execution Options

#### Option A: Docker Deployment (PostgreSQL + Redis Included)

```
cp .env.example .env
docker compose up --build
```

Access points: Web Application at `http://localhost:5173` | Interactive API Docs at `http://localhost:8000/docs`

### Option B: Infrastructure-Free Setup (SQLite + In-Memory Fallback)

```
# Backend Setup
cd backend
python -m venv .venv
.venv\Scripts\activate     # Linux/macOS: source .venv/bin/activate
pip install -r requirements.txt
python -m app.seed
uvicorn app.main:app --reload --port 8000

# Frontend Setup (Second Terminal)
cd frontend
npm install
npm run dev
```

## Role-Based Security & Redaction Matrix

Role authentication is enforced server-side via JWT tokens. Restricted metrics are pruned at the API boundary, guaranteeing zero leakage to the UI client.

| Username / Password | Role Persona | Data Visibility Scope | Approval Workflow Privileges |
|---|---|---|---|
| `executive` | Executive | All 5 KPIs, aggregated strategic views, customer data | Budget approvals, strategic initiatives, inventory reallocation |
| `marketing` | Marketing Manager | Revenue, Orders, Conversion Rate, AOV, Ad Spend | Campaign budget adjustments, ad creative levers |
| `operations` | Operations Manager | Revenue, Orders, Gross Margin, Cost, Supply Availability | Inventory re-orders, supplier SLA adjustments |
| `analyst` | Data Analyst | Unfiltered telemetry, OLS formulas, $R^2$, $p$-values | Feedback submissions, hypothesis rating |

# 3. System Architecture & Technology Stack

```
Transactions ─┐
Marketing   ─┼─> Reconciliation -> Semantic Layer -> Movement Engine + Quality Check
External    ─┘                                                   |
                                                                 v
                          Statistics / ML / Business Rules -> Driver Attribution (OLS/LMDI)
                                                                 |
                                                                 v
                                         Evidence Layer -> Confidence Engine (Composite)
                                                             /
                                                         |                    |
                                           Persona Narrative (LLM)        ABSTAIN
                                                         |                    |
                                                 Action Recommendation        |
                                                         └─> Feedback <───────┘
                                                                 |
                                                          Evaluation Loop
```

## Layer Technology Mapping

| Layer | Technology | Engineering Purpose | Implementation File |
|---|---|---|---|
| Data Generation | Seeded Mulberry32 PRNG | Generates 365 days of multi-table synthetic enterprise data | `app/engine/generator.py` |
| Semantic Layer | KPI Contracts (Dict) | Single source of truth for definitions, formulas, grains, owners | `app/engine/semantic.py` |
| Movement Engine | Pandas, NumPy, Z-Score | Rolling baseline anomaly detection vs 84-day historical window | `app/engine/analytics.py` |
| Factor Decomposition | LMDI Additive Model | Exact mathematical decomposition (Orders vs AOV) with zero residual | `app/engine/analytics.py` |
| Driver Attribution | Statsmodels OLS | Multi-variate regression yielding beta coefficients, $R^2$, and $p$-values | `app/engine/analytics.py` |
| Confidence Engine | Weighted Composite (5 Signals) | Multi-factor quality scoring assessing completeness, freshness, agreement | `app/engine/analytics.py` |
| Narrative Validation | Regex + Pydantic Rules | Post-hoc verification rejecting any unverified numerical claim | `app/engine/narrative.py` |
| API Gateway | FastAPI + Pydantic | Async REST endpoints, OpenAPI auto-documentation, CORS handling | `app/main.py` |
| Data Ingestion | REST Endpoint (JSON/CSV) | Custom dataset ingestion ( `POST /api/ingest` ) & database seeding | `app/main.py` |

# 4. Mathematical Foundations & Core Algorithms

### 1. Logarithmic Mean Divisia Index (LMDI) Factor Decomposition

Revenue is decomposed exactly into volume (Orders) and price/mix (AOV) effects:

```
Δ Revenue = Δ Orders · L(w) + Δ AOV · L(w)
where L(x, y) = (x - y) / (ln(x) - ln(y))
```

Unlike Shapley value approximations, LMDI additive decomposition leaves **zero residual error** ($ ext{Residual} = 0.000000$), ensuring dashboard figures reconcile perfectly.

### 2. Ordinary Least Squares (OLS) Driver Attribution

Attributes performance shifts across advertising spend, price index changes, and supply stockouts:

```
Y_revenue = β_0 + β_1 · X_ad_spend + β_2 · X_price_idx + β_3 · X_stockout + ε
```

Outputs exact beta coefficients, $t$-statistics, $p$-values, and $R^2$ goodness-of-fit metrics.

# 5. Selective Abstention & Confidence Meter

**CONFIDENCE SIGNAL BREAKDOWN**

- **Data Completeness (25%)**: Missing record ratio
- **Data Freshness (20%)**: Hours since last pipeline refresh
- **Statistical Strength (20%)**: Regression $p$-value & $R^2$
- **Cross-Source Agreement (20%)**: Signal alignment
- **Historical Baseline (15%)**: Days of history accrued

**EXPLICIT ABSTENTION TRIGGERS**

- Historical history < 30 days for affected slice
- Cross-source agreement < 0.50 with stale marketing data
- Composite confidence score < 60%

**Abstention Behaviour**

When an abstention trigger fires, the engine **refuses to output a driver or recommendation**. Instead, it states what data is missing/unreliable and prompts the analyst with clarifying operational questions.

## 6. Benchmark Scenarios Catalog

| Scenario Name | Injected Operational Anomaly | Engine Output & Behavior |
|---|---|---|
| `Revenue Decline` | Marketing spend pullback (-30%) + North region stockout | Explains: Revenue down -15.7%, identifies Paid Search ad spend as top driver |
| `Margin Compression` | Revenue steady, supplier unit cost inflation (+12%) | Explains: Gross Margin down -8.2%, attributes to unit cost shift |
| `Contradictory Evidence` | Marketing feed 29h stale, 18% external records missing | **ABSTAINS** — Surfaces 55% confidence score & asks clarifying questions |
| `New Product Launch` | NovaBud Air item launched only 18 days ago | **ABSTAINS** — Cites insufficient historical baseline (< 30 days) |

## 7. The LLM Boundary & Numerical Claim Validation

The LLM is invoked only when live narrative mode is requested. When called, raw datasets are **never sent to the LLM prompt** — only structured mathematical evidence objects.

```
[LLM Output Generation]
         │
         ▼
[Regex Number Extraction] ──> Extracts all numeric tokens (e.g., "$3.1M", "-15.7%", "52,605")
         │
         ▼
[Evidence Traceability Check] ──> Verifies each token against computed python evidence dictionary
         │
    ┌────┴─────────────────────────┐
    ▼                              ▼
[All Validated]          [Unverified Figure Detected]
    │                              │
    ▼                              ▼
[Accept & Output]        [Reject & Fallback to Template]
```

## 8. Complete REST API Specifications

| HTTP Method | Route Endpoint | Input Parameters | Primary Purpose |
|---|---|---|---|
| `POST` | `/api/auth/token` | `username`, `password` | Authenticates user role and issues JWT bearer token |
| `GET` | `/api/insight` | `scenario`, `role`, `live` | Returns full KPI movements, LMDI, narrative, and confidence |
| `GET` | `/api/drivers` | `scenario`, `dim`, `region` | Returns OLS regression, correlations, and dimensional mix |
| `GET` | `/api/sources` | `scenario` | Returns data source freshness, row counts, and schema definitions |
| `GET` | `/api/semantic` | None | Returns KPI Semantic Layer contracts and business metadata |
| `POST` | `/api/chat` | `question`, `scenario` | Natural language query routing to structured insights |
| `POST` | `/api/ingest` | JSON data payload | Custom dataset ingestion for transactions, marketing & external data |
| `POST` | `/api/seed` | None | Seeds database with all 4 benchmark scenario datasets |
| `GET` | `/api/health` | None | Returns P95 latency, cache hit rate, LLM-avoidance %, abstention % |

## 9. Automated Test Verification

```
$ pytest -q
.......................................
39 passed in 1.42s
```

The test suite asserts ground-truth recovery across all scenarios:

- LMDI factor effects reconcile to total revenue change within < $1.00.
- Contradictory scenario triggers selective abstention citing stale marketing data.
- New product scenario triggers selective abstention on sparse history (18 days).
- 100% of numerical figures in generated prose trace to evidence objects.
- Zero NaN, Null, or Infinity values in any response stream.